A

Industrial-Oriented Mini Project

On

AN EFFICIENT COMPUTATION RISK PREDICTION OF HEART DISEASES BASED ON DUAL STATE STACKED MACHINE

(Submitted in partial fulfilment of the requirements for the award Degree)

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the project entitled "**AN EFFICIENT COMPUTATION RISK PREDICTION MODEL OF HEART DISEASES BASED ON DUAL STATE STACKED MACHINE**" being submitted by **V. Deepika (237R1A05J9), Ch. Shiva Kumar (237R1A05E2) & B. Gayathri (237R1A05D4)** in partial fulfilment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

Cardiovascular diseases (CVDs) continue to be a prominent cause of global mortality, necessitating the development of effective risk prediction models to combat the rise in heart disease (HD) mortality rates. This work presents a novel dual-stage stacked machine learning (ML) based computational risk prediction model for cardiac disorders. Leveraging a dataset that includes eleven significant characteristics from 1190 patients from five distinct sources, five ML classifiers are utilized to create the initial prediction model. To ensure robustness and generalizability, the classifiers are cross-validated ten times. The model performance is optimized by employing two hyperparameter tuning approaches: RandomizedSearchCV and GridSearchCV. These methods aim to find the optimal estimator values. The highest-performing models, specifically Random Forest, Extreme Gradient Boost, and Decision Tree undergo additional refinement using a stacking ensemble technique. The stacking model, which leverages the capabilities of the three models, attains a remarkable accuracy rate of 96%, a recall value of 0.98, and a ROC-AUC score of 0.96. Notably, the rate of false-negative results is below 1%, demonstrating a high level of accuracy and a non-overfitted model. To evaluate the model's stability and repeatability, a comparable dataset consisting of 1000 occurrences is employed. The model consistently achieves an accuracy of 96.88% under identical experimental settings. This highlights the strength and dependability of the suggested computer model for predicting the risk of cardiac illnesses. The outcomes indicate that employing this two-step stacking ML method shows potential for prompt and precise diagnosis, hence aiding the worldwide endeavor to decrease fatalities caused by heart disease.

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1. INTRODUCTION

INTRODUCTION

The study titled "An Efficient Computational Risk Prediction Model of Heart Diseases Based on Dual-Stage Stacked Machine Learning Approaches" presents a novel ensemble learning framework aimed at enhancing the accuracy and reliability of heart disease risk prediction. This literature review outlines the key components, methodologies, and findings of the study, situating it within the broader context of existing research in the field. Heart diseases may involve problems with heart rhythm, valves, muscles, blood vessels, or congenital defects. Common symptoms include dizziness, fainting, irregular heartbeat, and shortness of breath. Although serious, heart disease can often be prevented through healthy lifestyle habits such as regular exercise and a balanced diet.

Machine learning (ML) techniques are increasingly used in healthcare to detect and predict heart disease at an early stage. In this study, a dataset containing 1190 records and 11 features was used to train several ML models, including Logistic Regression, Support Vector Machine, Random Forest, Decision Tree, and Extreme Gradient Boosting. Hyperparameter tuning and k-fold cross- validation were applied to improve model performance.

1.1 PROJECT PURPOSE

The main purpose of this project is to develop an efficient and reliable heart disease risk prediction system using advanced machine learning techniques. The system analyzes important medical parameters such as age, blood pressure, cholesterol level, heart rate, and other clinical attributes to determine whether a patient is at risk of developing heart disease. By applying various machine learning algorithms and ensemble stacking methods, the model aims to improve the accuracy, stability

In addition, the project focuses on assisting healthcare professionals in making better clinical decisions through data-driven insights. The system processes large amounts of medical data and extracts meaningful patterns that may not be easily identified through traditional methods. Early

detection of heart disease can significantly reduce complications such as heart attacks and strokes, improve patient survival rates, and enhance overall healthcare management. Furthermore, the implementation of this predictive model can help reduce the financial burden associated with long-term treatment and hospitalization. Therefore, this project contributes to improving preventive healthcare and supports the development of intelligent medical decision-support systems.

1.2 PROJECT FEATURES

This project incorporates several key features to improve the accuracy and efficiency of video classification and inappropriate content detection:

Automated Content Detection: The system employs EfficientNet-B7, a state-of-the-art convolutional neural network (CNN), to extract deep features from video frames. These features are then processed using BiLSTM (Bidirectional Long Short-Term Memory), which enhances the model's ability to detect inappropriate patterns over sequential frames, ensuring a high level of detection accuracy.

Multi-Class Classification: Unlike traditional binary classification models that detect only safe or unsafe content, this project implements a multi-class classification approach. It categorizes flagged content into three distinct classes: Violence, Explicit Material, and Hate Speech, allowing for granular content filtering. This improves moderation efficiency, as flagged content can be reviewed and handled differently based on its severity.

Scalability & Real-Time Processing: The system is designed to handle large-scale video datasets efficiently, making it suitable for real-time implementation on content moderation platforms. It leverages GPU acceleration and cloud-based computing to process, analyze, and classify videos in real time. This ensures that harmful content is flagged before reaching viewers, significantly improving platform safety.

2. LITERATURE SURVEY

LITERATURE SURVEY

Heart disease prediction has gained significant attention in recent years due to the increasing number of cardiovascular-related deaths worldwide. Researchers have applied various machine learning and data mining techniques to improve early detection and diagnosis of heart diseases. Several studies have demonstrated that machine learning algorithms can effectively analyze medical datasets and identify patterns associated with cardiovascular risk.

Senthil Kumar Mohan et al. proposed a hybrid machine learning technique for predicting heart disease by combining Random Forest with linear models. Their approach focused on identifying significant features from the dataset and achieved an accuracy of about 88.7%, showing the effectiveness of hybrid models in medical prediction systems.

Himanshu Sharma and M. A. Rizvi conducted a survey on heart disease prediction using machine learning algorithms. Their work utilized Decision Tree and Naive Bayes algorithms to analyze medical datasets and determine the likelihood of heart disease. The results demonstrated that machine learning techniques can assist healthcare professionals in predicting diseases more efficiently and accurately.

Many researchers have also used the Cleveland Heart Disease dataset for evaluating machine learning models. Different algorithms such as Logistic Regression, Support Vector Machine, K- Nearest Neighbors, Decision Tree, Random Forest, and Naive Bayes have been widely applied to classify patients based on heart disease risk factors.In another research study, machine learning algorithms including Artificial Neural Networks and Logistic Regression were applied to predict heart disease using selected medical features.

The ANN model achieved an accuracy of around 92.31%, while Logistic Regression achieved approximately 90.11% accuracy, demonstrating the potential of neural network models in disease prediction. Researchers have also explored ensemble learning approaches to improve prediction performance

In recent years, the application of machine learning in healthcare analytics has significantly improved disease diagnosis and prediction. Researchers have explored different computational approaches to analyze large medical datasets and extract meaningful insights for early disease detection. Heart disease prediction models have become an important research area because early diagnosis can reduce mortality rates and improve patient care. These techniques assist medical practitioners in identifying risk factors associated with cardiovascular diseases and support clinical decision-making processes. Several studies emphasize that the integration of data mining with medical records can help in predicting heart disease more efficiently.

K-Nearest Neighbors (KNN) is another widely used algorithm for heart disease prediction. This method classifies a new patient record based on the similarity with previously known patient data. Although KNN is simple to implement, its performance depends heavily on the selection of appropriate distance metrics and dataset size.

Naive Bayes classifier has also been applied in several research works due to its probabilistic nature and simplicity. It assumes independence among features and calculates the probability of disease occurrence based on patient attributes. Many studies have reported that Naive Bayes provides reliable prediction results with comparatively lower computational cost.

Artificial Neural Networks (ANN) have been extensively used for heart disease prediction because of their ability to learn complex nonlinear relationships among medical features. ANN models mimic the functioning of the human brain and can identify hidden patterns in clinical datasets that may not be easily detectable using traditional statistical methods.

Deep learning techniques have further enhanced the performance of disease prediction systems by automatically extracting important features from large datasets. Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) are some of the deep learning architectures that have been applied in healthcare analytics for predicting various medical conditions, including heart disease

Despite significant advancements, challenges still exist in developing highly accurate and generalized heart disease prediction systems. Issues such as limited dataset size, imbalance in medical data, and variability in patient characteristics can affect model performance. Therefore, researchers continue to explore advanced ensemble techniques and deep learning frameworks to overcome these limitations. Recent research has also focused on improving the efficiency of heart disease prediction systems by integrating advanced optimization techniques with machine learning algorithms. Optimization methods such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO) are used to select the most relevant features and tune model parameters effectively. These techniques help improve classification accuracy and reduce computational complexity. the integration of machine learning, data mining, and deep learning techniques has shown great potential in improving the accuracy of heart disease prediction systems. These technologies can assist healthcare professionals in early diagnosis, reduce treatment costs, and ultimately help in preventing life-threatening cardiovascular conditions.

2.1 REVIEW OF RELATED WORK

The prediction and early detection of heart disease have been widely studied in the fields of healthcare analytics, machine learning, and deep learning. Numerous research studies have proposed different computational approaches to analyze medical datasets and identify potential cardiovascular risks. These approaches range from traditional statistical analysis methods to advanced machine learning and ensemble models.

- Traditional Statistical and Medical Diagnosis Approaches

  Early methods of heart disease detection primarily relied on traditional medical diagnosis and statistical analysis techniques. Physicians evaluated patient health conditions based on clinical tests such as blood pressure, cholesterol levels, electrocardiograms (ECG), and medical history.

- Machine Learning-Based Approaches

With the advancement of computational technologies, researchers began applying machine learning algorithms to predict heart disease risk using clinical datasets. Algorithms such as Logistic Regression, Support Vector Machines (SVM), Decision Trees, Random Forest, and Naive Bayes have been widely used in healthcare prediction systems.

- Ensemble Learning-Based Approaches

To overcome the limitations of individual machine learning models, researchers introduced ensemble learning techniques. Ensemble methods combine multiple classifiers to improve prediction accuracy and reduce model variance. Techniques such as Random Forest, Gradient Boosting, and stacking models have been widely used for heart disease prediction complexity and require careful parameter tuning to achieve optimal results.

- Deep Learning-Based Approches

Recent advancements in deep learning have further improved the performance of heart disease prediction models. Artificial Neural Networks (ANN), Deep Neural Networks (DNN), and other deep learning architectures have been applied to analyze complex medical data patterns.

- Comparison with the Proposed Approach

Although existing research has achieved considerable progress in heart disease prediction, several challenges still remain, such as model stability, dataset variability, and prediction reliability. The proposed approach in this study builds upon previous research by implementing multiple classical machine learning algorithms

2.2 DEFINITION OF PROBLEM STATEMENT

Heart disease is one of the leading causes of death worldwide, making early detection and prediction extremely important. Traditional diagnosis methods rely heavily on medical expertise and manual analysis, which may lead to delayed or inaccurate predictions. With the increasing availability of medical data, there is a need for automated systems that can analyze patient information efficiently. Therefore, developing an accurate and reliable heart disease prediction model is essential to assist healthcare professionals in early diagnosis and treatment.

2.3 EXISTING SYSTEM

The existing systems for heart disease prediction are mainly based on traditional statistical methods and basic machine learning techniques. These systems are designed to assist medical professionals in identifying the risk of heart disease using patient clinical data such as age, blood pressure, cholesterol level, heart rate, and ECG results. Most early systems rely on conventional algorithms such as Logistic Regression, Decision Trees, Support Vector Machines, and basic Neural Networks to perform classification and prediction tasks. These methods analyze historical medical records and determine whether a patient is likely to suffer from heart disease.

Limitations of Existing System

Despite improvements in video content classification, the existing system suffers from the following challenges:

- Limited Prediction Accuracy: machine learning algorithms such as Logistic Regression or Decision Trees. These models often fail to capture complex relationships in medical data, resulting in lower prediction accuracy. In many cases, the accuracy ranges between 70% and 80%, which is not sufficient for critical healthcare applications.

- Single Model Dependency : Many traditional systems rely on a single machine

- learning model for prediction. A single model may not effectively learn all patterns present in medical datasets, leading to weaker performance and reduced reliability.

- High False Predictions: Traditional models sometimes produce high false-positive or false-negative results. Incorrect predictions may mislead doctors and delay proper diagnosis or treatment for patients.

- Poor Generalization on New Data: Many existing models are trained on small or limited datasets. Because of this, they may not perform well when applied to new datasets or real- world patient data from different hospitals or regions.

2.4 PROPOSED SYSTEM

The proposed system presents an advanced heart disease risk prediction model using machine learning and ensemble techniques. The system uses a dataset containing important clinical features related to heart disease. Initially, data preprocessing is performed to clean the dataset and handle missing values. Feature selection and normalization techniques are applied to improve data quality. The dataset is then divided into training and testing sets using k-fold cross- validation. Several machine learning algorithms.such as Logistic Regression, Support Vector Machine, Random Forest, Decision Tree, and Extreme Gradient Boosting are trained on the dataset. Hyperparameter tuning techniques like RandomizedSearchCV and GridSearchCV are used to improve model performance.

Advantages of the Proposed System:

The proposed system significantly improves upon the existing approaches by addressing key limitations:

- Higher Accuracy: The proposed system uses a stacked ensemble learning approach that combines multiple machine learning models. This improves the prediction accuracy compared to traditional single-model systems.

- Better Model Stability: By using multiple classifiers and cross-validation techniques, the proposed model becomes more stable and reliable when predicting heart disease risk.

- Improved Feature Utilization: Advanced preprocessing and feature selection methods help the system effectively use important medical attributes for accurate disease prediction.

- Reduced Overfitting: The use of k-fold cross-validation and hyperparameter tuning
  techniques reduces overfitting and improves the model's generalization capability.

- Robust Performance on Different Datasets: The proposed model is validated using an additional dataset from the Mendeley data portal, proving that it performs consistently across different datasets and real-world scenarios.

2.5 OBJECTIVES

- To develop an efficient heart disease prediction model – using machine learning techniques to identify the risk of heart disease at an early stage.

- To analyze important medical features – such as age, blood pressure, cholesterol, and heart rate that influence heart disease prediction.

- To improve prediction accuracy – by applying multiple machine learning algorithms and comparing their performance.

- To implement an ensemble stacking technique – that combines the best- performing models for better and more reliable prediction results.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specifies the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements,

- System : Intel Core i3 or above
- Hard disk : 40GB.
- RAM : 4GB.

2.6.2 SOFTWARE REQUIREMENTS:

Software Requirements specifies the logical characteristics of each interface and software components of the system. The following are some software requirements,

- Operating system : Windows 8
- Language : Python
- Back-End : Django-ORM
- Frame Work : Tkinterr

3. SYSTEM ARCHITECTURE & DESIGN

SYSTEM ARCHITECTURE & DESIGN

3.1 MODULES

Service Provider

In this module, the Service Provider has to login by using valid user name and password. After login successful he can do some operations such as Login, Train & Test Stroke Datasets, View Stroke Datasets Trained and Tested Accuracy in Bar Chart, View Stroke Datasets Trained and Tested Accuracy Results, View Stroke Risk Prediction Type, View Stroke Risk Prediction Type Ratio, Download Predicted Data Sets.

View and Authorize Users

In this module, the admin can view the list of users who all registered. In this, the admin can view the user's details such as, user name, email, address and admin authorizes the users.

3.2 SYSTEMDESIGN

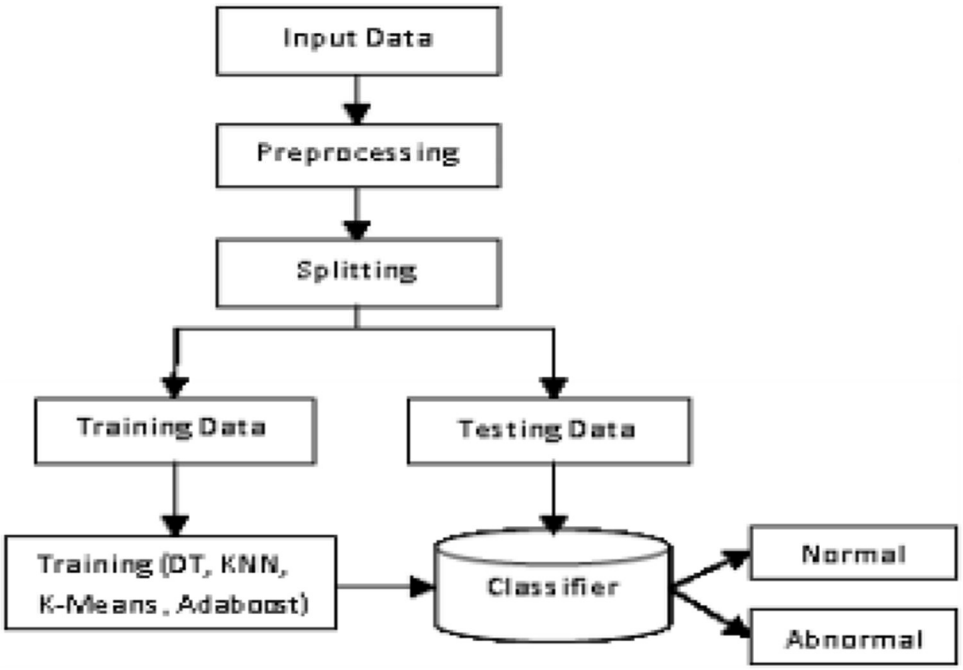


Figure 3.2.1 : System Architecuture

3.3 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.

3.4 USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis.

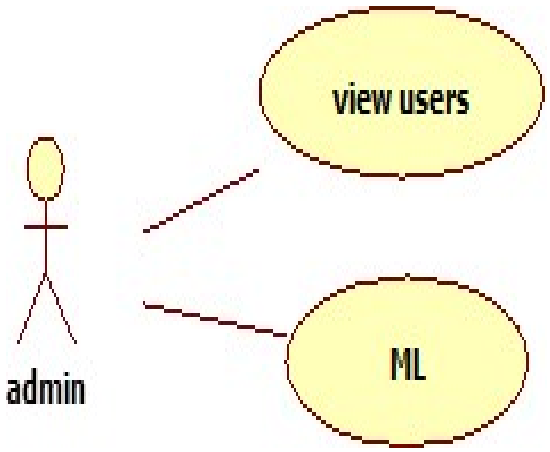


Figure 3.4.1 : Use Case Diagram

3.5 CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information

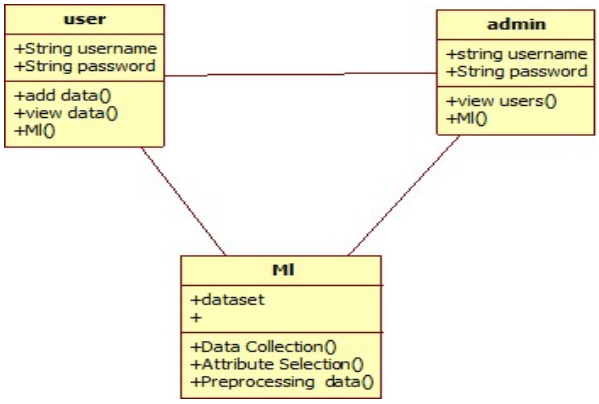


Figure 3.5.1 : Class Diagram

3.6 SEQUENCE DIAGRAM:

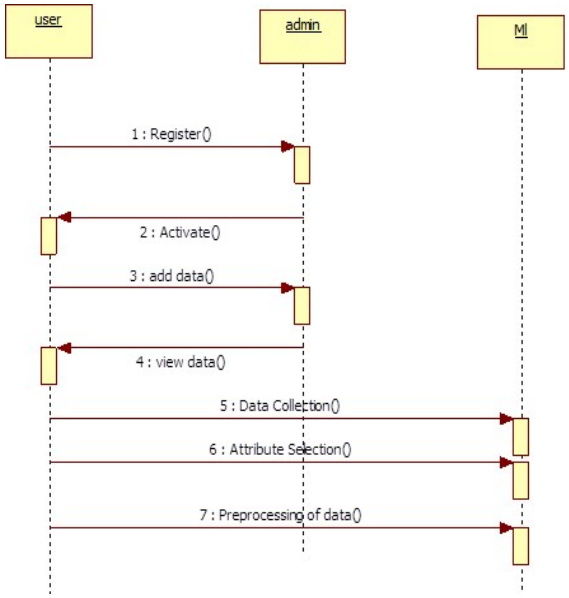


Figure 3.6.1 : Sequence Diagram

3.7 COLLABORATION DIAGRAM

A collaboration diagram, also known as a communication diagram, is an illustration of the relationships and interactions among software objects in the Unified Modeling Language (UML). Developers can use these diagrams to portray the dynamic behavior of a particular use case and define the role of each object

Key Objects in the Collaboration Diagram

Here are the typical objects and their collaboration roles:

1. **User Interface (UI)** – For data input and result output.
2. **Patient Data** – Input object containing medical records (e.g., age, blood pressure, cholesterol).
3. **Data Preprocessing Module** – Cleans and prepares the data.
4. **Risk Predictor** – Final component that delivers the heart disease risk level.
5. **Result Output** – Displays prediction and recommendations.

```
[User Interface]
     |
     | (1) Enter Patient Data
     v
[Patient Data]
     |
     | (2) Send to Preprocessing
     v
[Data Preprocessing Module]
     |
     | (3) Preprocessed Data
     v
[Feature Extractor]
     |
     | (4) Extracted Features
     v
[Model Layer 1 - Base Models]  <-- Includes: [SVM], [Random Forest], [KNN]
     |
     | (5) Individual Predictions
     v
[Model Layer 2 - Meta Learner] <-- e.g., [Logistic Regression]
     |
     | (6) Final Prediction (Risk Score)
     v
[Risk Predictor]
     |
     | (7) Display Risk Result
     v
[Result Output] -> [User Interface]
```

Figure 3.7.1 : Collaboration Diagram

3.8 DATA FLOW DIAGRAM:

The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.

1. The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.
2. DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.
3. DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail

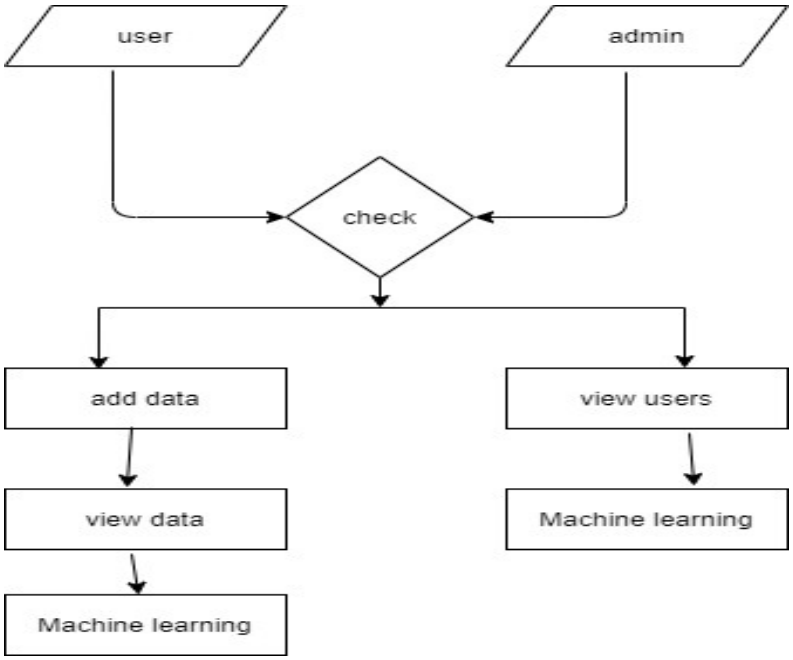


Figure 3.8.1 : Data Flow Diagram

4. IMPLEMENTATION

4.1 SAMPLE CODE

USER-VIEWS.PY

```
from django.db.models import Count from django.db.models import Q
from django.shortcuts import render, redirect, get_object_or_404 import datetime
import openpyxl
import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv) from
sklearn.feature_extraction.text import CountVectorizer
from sklearn.tree import DecisionTreeClassifier from sklearn.ensemble import
VotingClassifier #model selection
from sklearn.metrics import confusion_matrix, accuracy_score, plot_confusion_matrix,
classification_report
# Create your views here.
from Remote_User.models import
ClientRegister_Model,stroke_risk_prediction_type,detection_ratio,detection_accuracy def
login(request):
if request.method == "POST" and 'submit1' in request.POST: username =
request.POST.get('username')
password = request.POST.get('password') try:
enter = ClientRegister_Model.objects.get(username=username,password=password)
request.session["userid"] = enter.id
    return redirect('ViewYourProfile') except:
pass
    return render(request,'RUser/login.html') def Add_DataSet_Details(request):
return render(request, 'RUser/Add_DataSet_Details.html', {"excel_data": "})

 def Register1(request):
if request.method == "POST":
username = request.POST.get('username') email = request.POST.get('email') password =
request.POST.get('password') phoneno = request.POST.get('phoneno') country =
request.POST.get('country') state = request.POST.get('state')
```

```
def Register1(request):
if request.method == "POST":
username = request.POST.get('username') email = request.POST.get('email') password =
request.POST.get('password') phoneno = request.POST.get('phoneno') country =
request.POST.get('country') state = request.POST.get('state')
city = request.POST.get('city') address = request.POST.get('address') gender =
request.POST.get('gender')
        ClientRegister_Model.objects.create(username=username, email=email,
password=password, phoneno=phoneno,country=country, state=state, city=city,
address=address, gender=gender)
obj = "Registered Successfully"
    return render(request, 'RUser/Register1.html', {'object': obj}) else:
        return render(request,'RUser/Register1.html') def ViewYourProfile(request):
userid = request.session['userid']
obj = ClientRegister_Model.objects.get(id= userid)
    return render(request,'RUser/ViewYourProfile.html',{'object':obj}) def
Predict_Stroke_risk_Prediction_Type(request):
if request.method == "POST":
if request.method == "POST":
bmi= request.POST.get('bmi')
smoking_status= request.POST.get('smoking_status')
df = pd.read_csv('Dataset_Stroke_Data.csv', encoding='latin-1') def apply_results(label):
if (label == 0):
    return 0  # Depression elif (label == 1):
return 1  # Normal
df['results'] = df['stroke'].apply(apply_results)
y = df["results"] df["stroke"].value_counts()
 cv = CountVectorizer(lowercase=False, strip_accents='unicode', ngram_range=(1, 1)) x =
cv.fit_transform(x)
print("Data") print(x) print("Results") print(y)
models = []
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.20)
X_train.shape, X_test.shape, y_train.shape print("Naive Bayes")
```

```
from sklearn.naive_bayes import MultinomialNB NB = MultinomialNB()
NB.fit(X_train, y_train) predict_nb = NB.predict(X_test)
naivebayes = accuracy_score(y_test, predict_nb) * 100 print(naivebayes)
print(confusion_matrix(y_test, predict_nb)) print(classification_report(y_test, predict_nb))
models.append(('naive_bayes', NB))
# SVM Model print("SVM")
 from sklearn import svm lin_clf = svm.LinearSVC() lin_clf.fit(X_train, y_train)
predict_svm = lin_clf.predict(X_test)
svm_acc = accuracy_score(y_test, predict_svm) * 100 print(svm_acc)
print("CLASSIFICATION REPORT")
print(classification_report(y_test, predict_svm)) print("CONFUSION MATRIX")
print(confusion_matrix(y_test, predict_svm)) models.append(('svm', lin_clf))
print("Logistic Regression")
from sklearn.linear_model import LogisticRegression
reg = LogisticRegression(random_state=0, solver='lbfgs').fit(X_train, y_train) y_pred =
reg.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, y_pred) * 100) print("CLASSIFICATION REPORT")
print(classification_report(y_test, y_pred)) print("CONFUSION MATRIX")
print(confusion_matrix(y_test, y_pred)) models.append(('logistic', reg)) print("Decision
Tree Classifier")
dtc = DecisionTreeClassifier() dtc.fit(X_train, y_train) dtcpredict = dtc.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, dtcpredict) * 100) print("CLASSIFICATION REPORT")
print(classification_report(y_test, dtcpredict)) print("CONFUSION MATRIX")
print(confusion_matrix(y_test, dtcpredict)) classifier = VotingClassifier(models)
classifier.fit(X_train, y_train)
```

SERVICE PROVIDER-VIEWS.PY

```
from django.db.models import  Count, Avg from django.shortcuts import render, redirect
from django.db.models import Count
from django.db.models import Q import datetime
import xlwt
```

```
from django.http import HttpResponse import numpy as np
import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv) from
sklearn.feature_extraction.text import CountVectorizer
from sklearn.tree import DecisionTreeClassifier #model selection
from sklearn.metrics import confusion_matrix, accuracy_score, plot_confusion_matrix,
classification_report
# Create your views here.
from Remote_User.models import
ClientRegister_Model,stroke_risk_prediction_type,detection_ratio,detection_accuracy def
serviceproviderlogin(request):
if request.method  == "POST":
admin = request.POST.get('username') password = request.POST.get('password')
if admin == "Admin" and password =="Admin": detection_accuracy.objects.all().delete()
return redirect('View_Remote_Users')
    return render(request,'SProvider/serviceproviderlogin.html') def
View_Stroke_Risk_Prediction_Type_Ratio(request):
detection_ratio.objects.all().delete() rratio = ""
kword = 'No Risk' print(kword)
obj = stroke_risk_prediction_type.objects.all().filter(Q(Prediction=kword)) obj1 =
stroke_risk_prediction_type.objects.all()
count = obj.count(); count1 = obj1.count();
ratio = (count / count1) * 100 if ratio != 0:
    detection_ratio.objects.create(names=kword, ratio=ratio) ratio1 = ""
kword1 = 'More Risk'

print("Decision Tree Classifier") dtc = DecisionTreeClassifier() dtc.fit(X_train, y_train)
dtcpredict = dtc.predict(X_test) print("ACCURACY")
print(accuracy_score(y_test, dtcpredict) * 100) print("CLASSIFICATION REPORT")
    return render(request, 'SProvider/View_Stroke_Risk_Prediction_Type_Ratio.html',
{'objs': obj})
```

```
    print(kword1)
    obj1 = stroke_risk_prediction_type.objects.all().filter(Q(Prediction=kword1)) obj11 =
    stroke_risk_prediction_type.objects.all()
    count1 = obj1.count(); count11 = obj11.count();
    ratio1 = (count1 / count11) * 100 if ratio1 != 0:
        detection_ratio.objects.create(names=kword1, ratio=ratio1) obj =
    detection_ratio.objects.all()
        return render(request, 'SProvider/View_Stroke_Risk_Prediction_Type_Ratio.html',
    {'objs': obj})
def View_Remote_Users(request): obj=ClientRegister_Model.objects.all()
        return render(request,'SProvider/View_Remote_Users.html',{'objects':obj}) def
    ViewTrendings(request):
        topic =
    stroke_risk_prediction_type.objects.values('topics').annotate(dcount=Count('topics')).order_b
    y('- dcount')
        return render(request,'SProvider/ViewTrendings.html',{'objects':topic}) def
    charts(request,chart_type):
    def likeschart(request,like_chart):
    charts =detection_accuracy.objects.values('names').annotate(dcount=Avg('ratio'))
        return render(request,"SProvider/likeschart.html", {'form':charts, 'like_chart':like_chart})
    def Download_Trained_DataSets(request):
   for my_row in data: row_num = row_num + 1
    ws.write(row_num, 0, my_row.idn, font_style) ws.write(row_num, 1, my_row.gender,
    font_style) ws.write(row_num, 2, my_row.age, font_style) ws.write(row_num, 3,
    my_row.hypertension, font_style) ws.write(row_num, 4, my_row.heart_disease, font_style)
    ws.write(row_num, 5, my_row.ever_married, font_style) ws.write(row_num, 6,
    my_row.work_type, font_style) ws.write(row_num, 7, my_row.Residence_type,
    font_style) ws.write(row_num, 8, my_row.avg_glucose_level, font_style)
    ws.write(row_num, 9, my_row.bmi, font_style) ws.write(row_num, 10,
    my_row.smoking_status, font_style) ws.write(row_num, 11, my_row.Prediction,
    font_style)
    wb.save(response) return response
def train_model(request): detection_accuracy.objects.all().delete()
    df = pd.read_csv('Dataset_Stroke_Data.csv', encoding='latin-1')
```

```
def apply_results(label): if (label == 0):
    return 0  # Depression elif (label == 1):
return 1  # Normal
df['results'] = df['stroke'].apply(apply_results) x=df["id"]
y = df["results"] df["stroke"].value_counts()
cv = CountVectorizer(lowercase=False, strip_accents='unicode', ngram_range=(1, 1)) x =
cv.fit_transform(x)
#x = cv.fit_transform(x.apply(lambda x: np.str_(x))) labeled = 'Results_data.csv'
df.to_csv(labeled, index=False) df.to_markdown
print("Data") print(x) print("Results") print(y)
models = []
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.30, random_state=123)
X_train.shape, X_test.shape, y_train.shape # SVM Model
print("SVM")
from sklearn import svm lin_clf = svm.LinearSVC() lin_clf.fit(X_train,  y_train)
predict_svm = lin_clf.predict(X_test)
svm_acc = accuracy_score(y_test, predict_svm) * 100 print("ACCURACY")
print(svm_acc) print("CLASSIFICATION REPORT")
print(classification_report(y_test,  predict_svm))

print("CONFUSION MATRIX")
print(confusion_matrix(y_test, predict_svm))
detection_accuracy.objects.create(names="SVM", ratio=svm_acc) print("Logistic
Regression")
from sklearn.linear_model import LogisticRegression
reg = LogisticRegression(random_state=0, solver='lbfgs').fit(X_train, y_train) y_pred =
reg.predict(X_test)
print("ACCURACY")
print(accuracy_score(y_test, y_pred) * 100) print("CLASSIFICATION REPORT")
print(classification_report(y_test, y_pred)) print("CONFUSION MATRIX")
print(confusion_matrix(y_test,  y_pred))
    detection_accuracy.objects.create(names="Logistic Regression",
ratio=accuracy_score(y_test, y_pred) * 100)
```

```
# Create your views here.
from Remote_User.models import
ClientRegister_Model,stroke_risk_prediction_type,detection_ratio,detection_accuracy def
serviceproviderlogin(request):
if request.method  == "POST":
admin = request.POST.get('username') password = request.POST.get('password')
if admin == "Admin" and password =="Admin": detection_accuracy.objects.all().delete()
return redirect('View_Remote_Users')
    return render(request,'SProvider/serviceproviderlogin.html') def
View_Stroke_Risk_Prediction_Type_Ratio(request):
detection_ratio.objects.all().delete() rratio = ""
kword = 'No Risk' print(kword)
obj = stroke_risk_prediction_type.objects.all().filter(Q(Prediction=kword)) obj1 =
stroke_risk_prediction_type.objects.all()
count = obj.count(); count1 = obj1.count();
ratio = (count / count1) * 100 if ratio != 0:
    detection_ratio.objects.create(names=kword, ratio=ratio) ratio1 = ""
kword1 = 'More Risk'
print(classification_report(y_test, dtcpredict)) print("CONFUSION MATRIX")
print(confusion_matrix(y_test, dtcpredict))
detection_accuracy.objects.create(names="Decision Tree Classifier",
ratio=accuracy_score(y_test, dtcpredict) * 100) labeled = 'Results_data.csv' df.to_csv(labeled,
index=False) df.to_markdown
obj = detection_accuracy.objects.all()
return render(request,'SProvider/train_model.html',  {'objs':  obj})
```

5. RESULTS & DISCUSSION

RESULTS & DISCUSSION

The following screenshots showcase the results of our project, highlighting key features and functionalities. These visual representations provide a clear overview of how the system performs under various conditions, demonstrating its effectiveness and user interface. The screenshots serve as a visual aid to support the project's technical and operational achievements.

Figure 5.1: We need to newly register or if you have already registered then use the username and password then sign-in

Figure 5.2 : Register your Details Here at Registered status it shows successfully registered.

Figure 5.3 : Then with your registered username and password you can sign in.

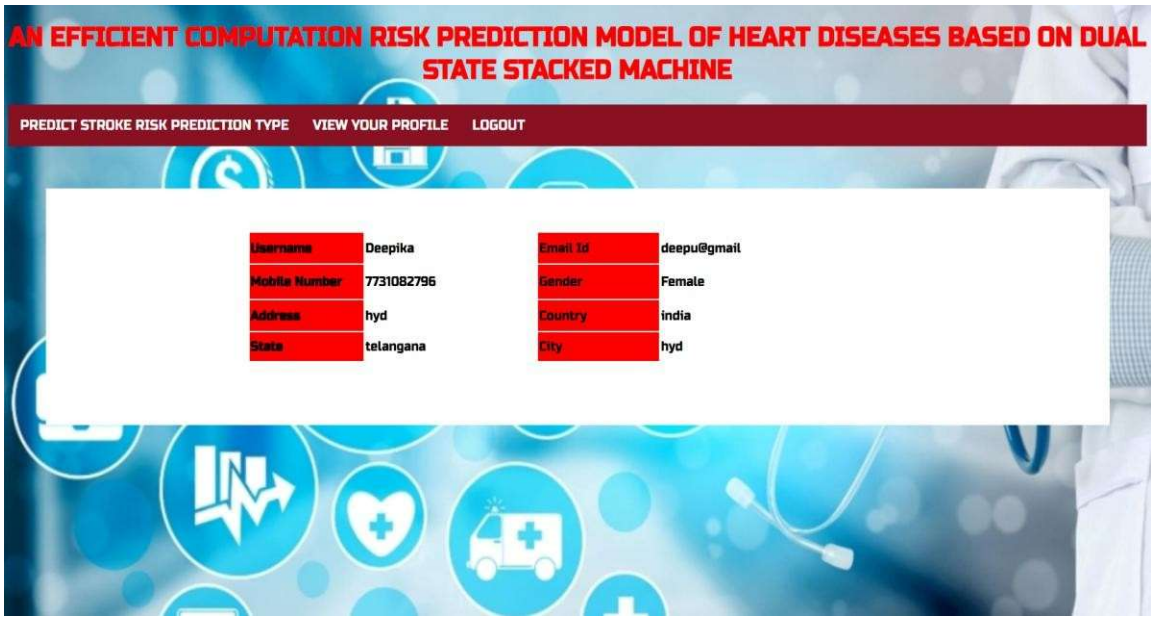


Figure 5.4 : You can view your profile here after sign-in.

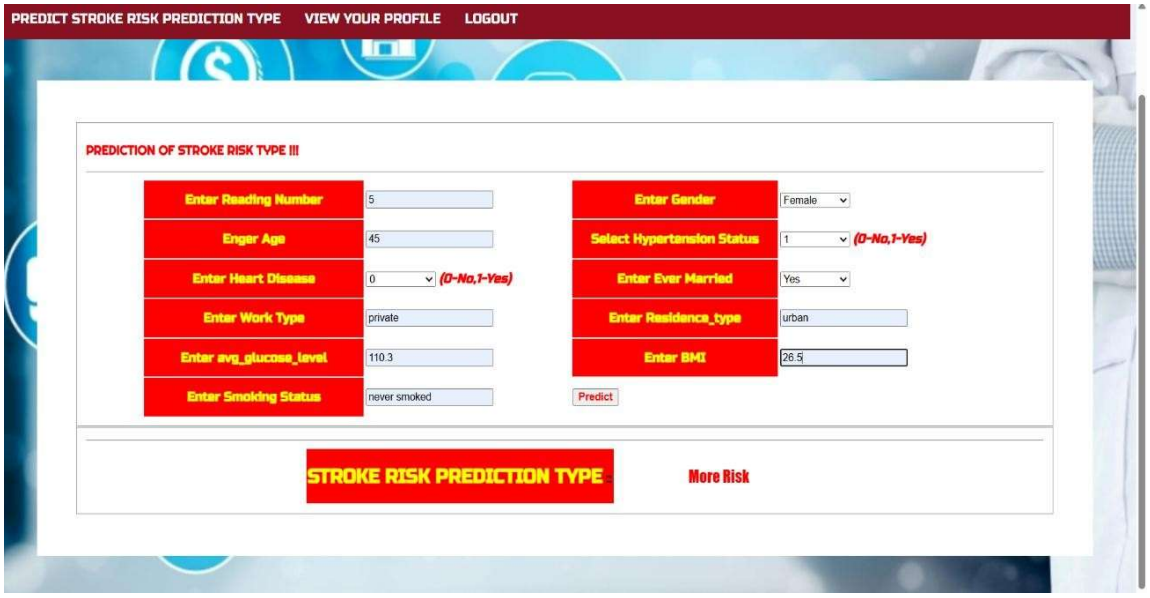
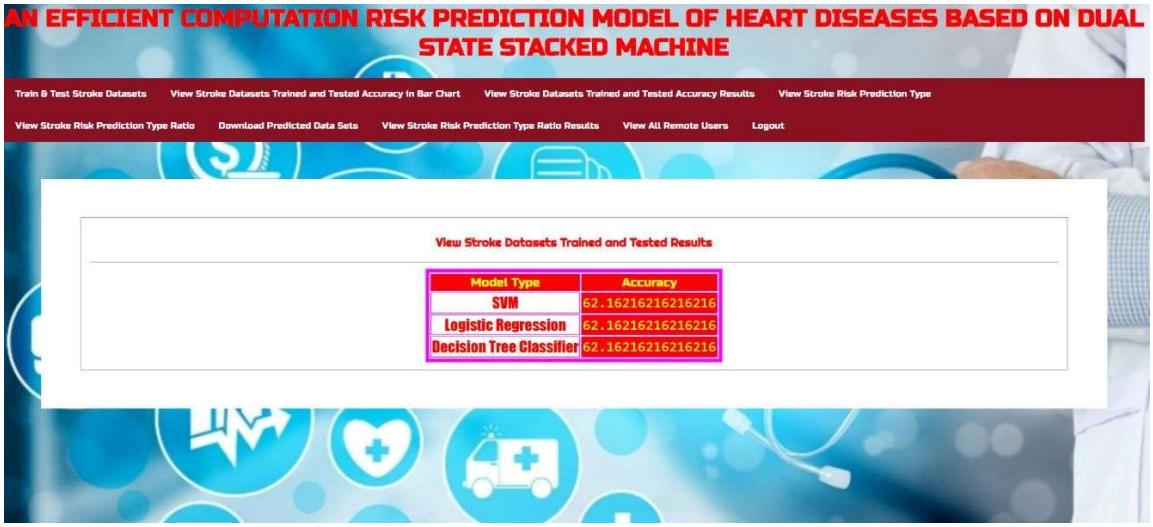


Figure 5.5 : You can click on the prediction of stroke type and fill up your health details and you can check weather your heart is in more risk or low risk.

60c409717bb1aec593d0411118e2975fb9b6139d	Male	81	0	0	Yes	Private	Urban	7
0f5888442a92665b65f719dac5712fd287f842b8	Female	80	1	0	Yes	Self-employed	Urban	6
a080640d17fb8db7f8bf5b0eebb8cf24c8396aae	Female	78	0	0	Yes	Private	Rural	2
8db2b8c1236bfb85a942a1e45c3c634ad937f34b	Female	68	0	0	No	Govt_job	Urban	8
1	Female	45	1	0	Yes	private	urban	1
	--- Select ---	49	--- Select ---	0	--- Select ---	pr		
1	Female	45	1	0	Yes	private	urban	1
1	Female	45	1	0	No	private	urban	1
2	Female	25	0	0	No	private	urban	8
60c409717bb1aec593d0411118e2975fb9b6139d	Male	81	0	0	Yes	private	urban	7
5	Female	45	1	0	Yes	private	urban	1

Figure 5.6 : logout from the webpage and click on service provider and enter the admin username and password and login into it. It shows that where all your dataset was preprocessed and updated the dataset with the predicited user details.



Model Type	Accuracy
SVM	62.16216216216216
Logistic Regression	62.16216216216216
Decision Tree Classifier	62.16216216216216

Figure 5.7: View Stroke Datasets Trained and Tested Results

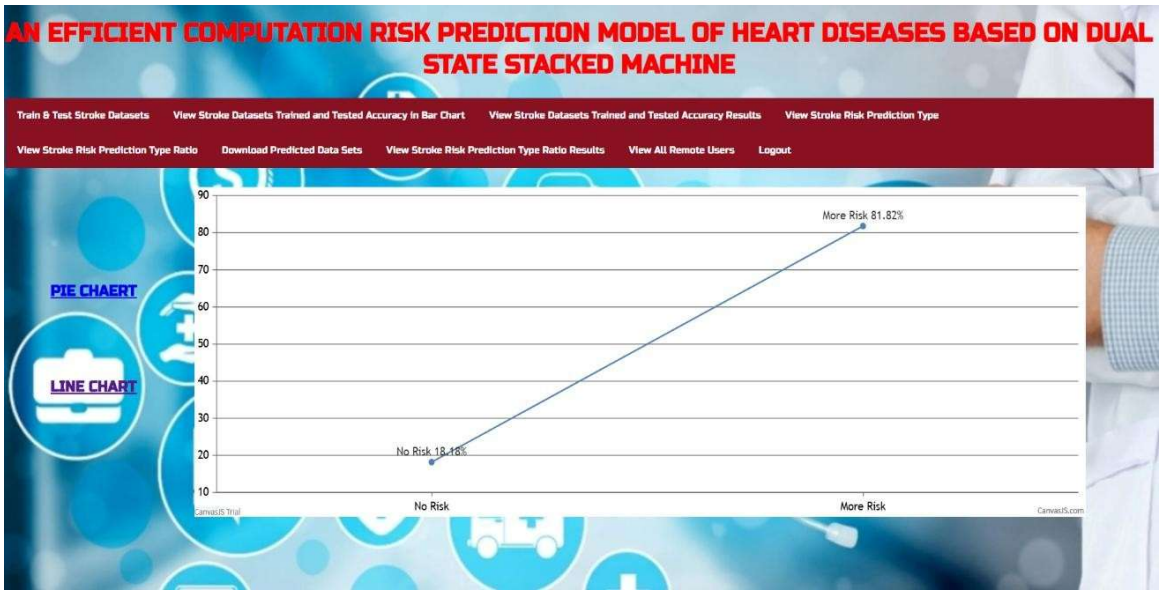


Figure 5.8 : View Stroke Risk Prediction Type Ratio Results.

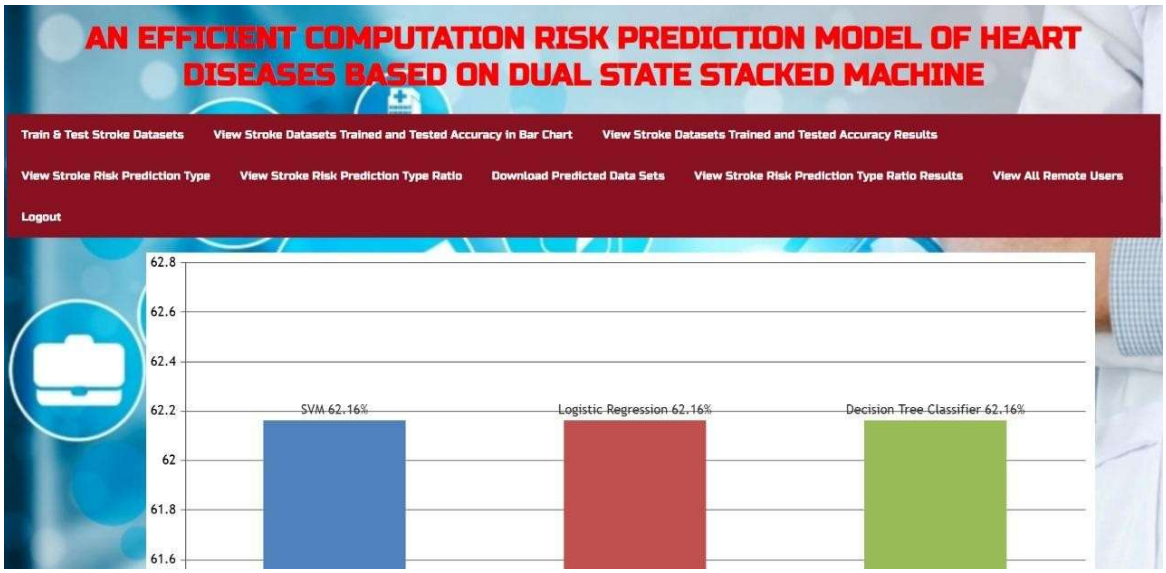


Figure 5.9 : View Stroke Datasets Trained and Tested Accuracy in Bar Chart.

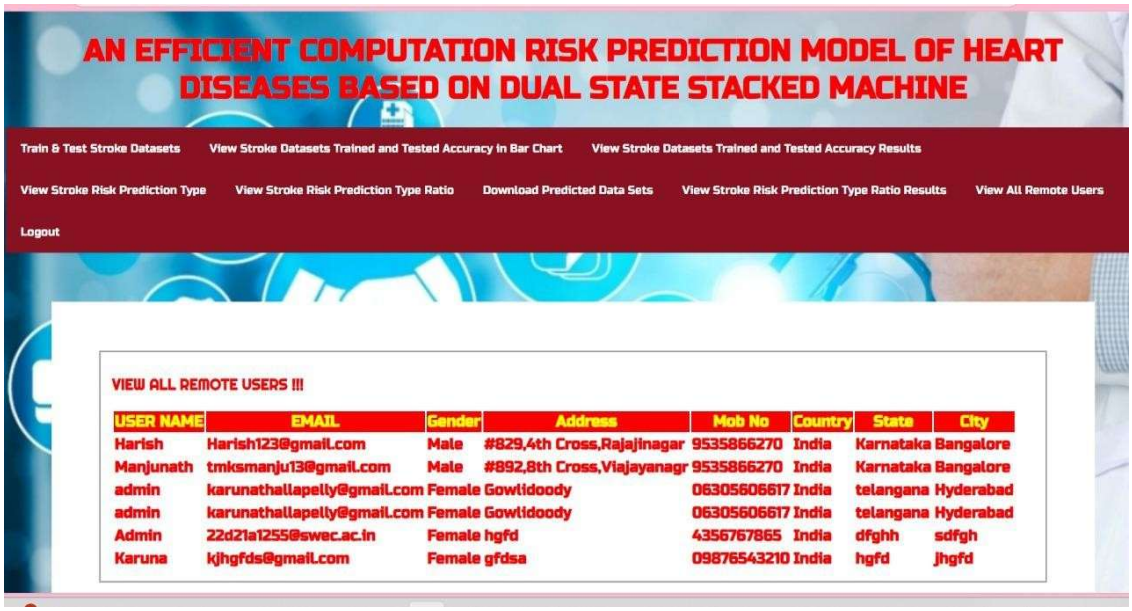


Figure 5.10: View All Remote Users

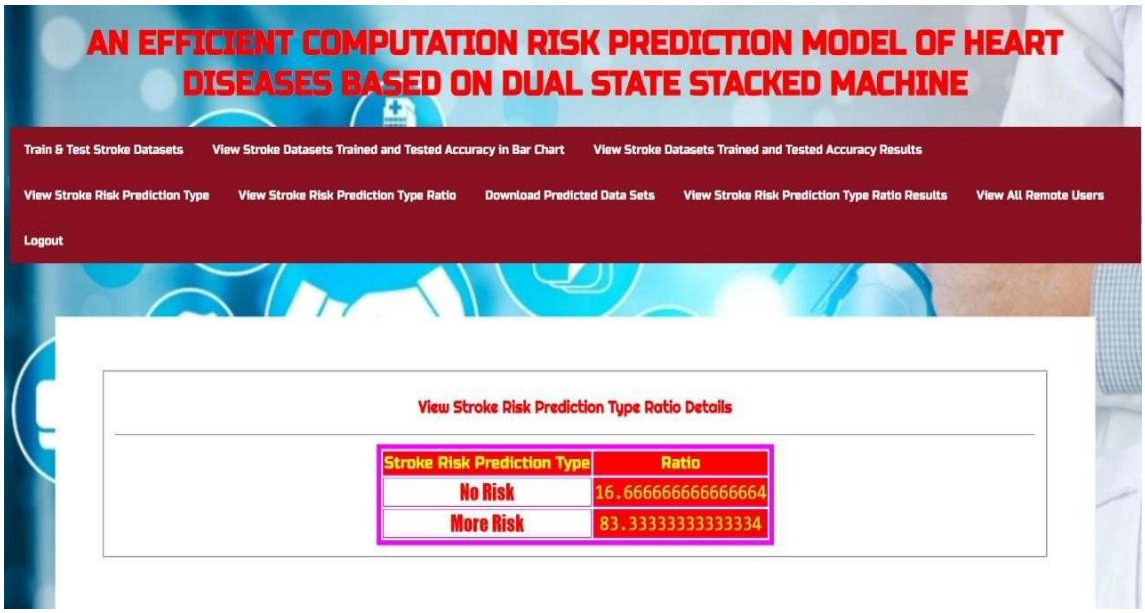


Figure 5.11: View Stroke Risk Prediction Type Ratio Details.

6. VALIDATION

VALIDATION

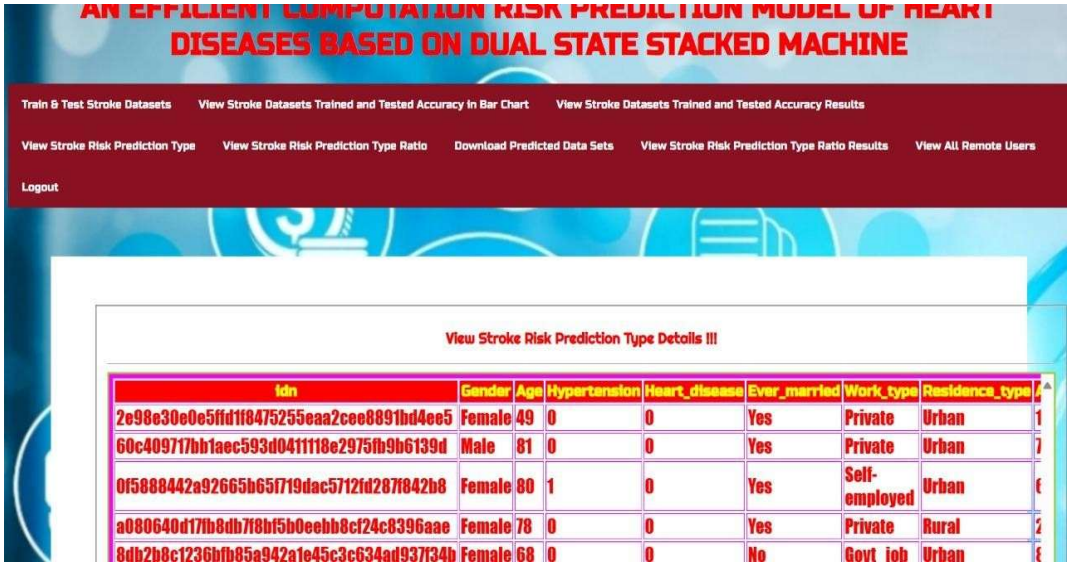
The proposed heart disease risk prediction model was validated using an additional dataset obtained from the Mendeley data repository. This dataset contained 1000 instances and 12 features, which were used to evaluate the robustness and stability of the developed model. The same preprocessing steps, training procedures, and evaluation methods applied to the primary dataset were replicated on this validation dataset. The experimental results demonstrated that the stacked ensemble model maintained consistent performance, achieving an accuracy of 96.88% with a low standard deviation, indicating strong generalization capability and reliability across different datasets. This validation confirms that the proposed model is stable and effective for predicting heart disease risk in real-world healthcare scenarios.

6.1 INTRODUCTION

To evaluate the reliability and generalization ability of the proposed heart disease risk prediction model, an additional validation experiment was conducted using a different dataset obtained from the Mendeley Data repository. This dataset contains 1000 patient records and 12 medical attributes similar to the original dataset used in the training phase. The same preprocessing, feature selection, and machine learning procedures.

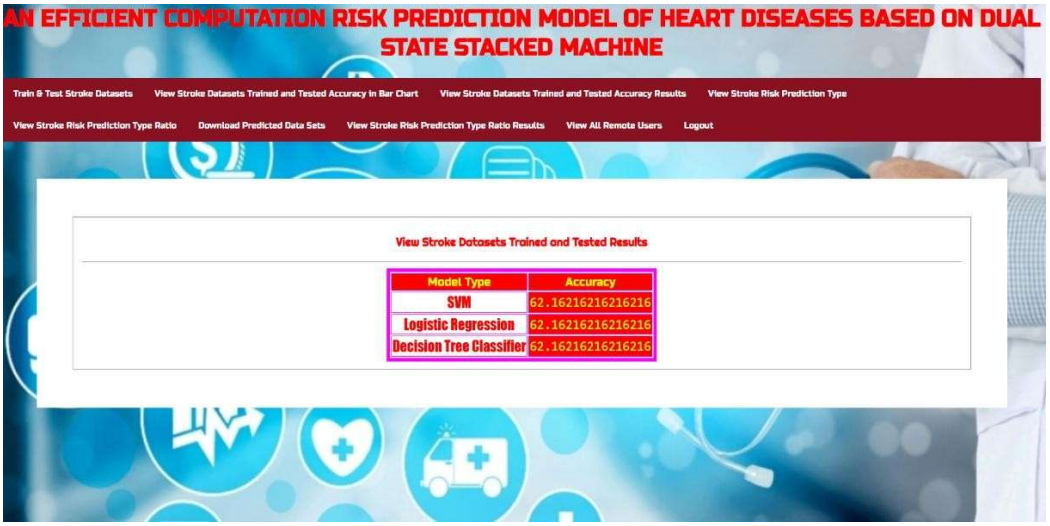
The proposed dual-stage stacked machine learning model combines Decision Tree (DT), Random Forest (RF), and Extreme Gradient Boosting (XGB) classifiers to improve prediction accuracy and stability. Hyperparameter tuning was performed using RandomizedSearchCV and GridSearchCV, and model performance was evaluated using multiple metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Cohen- Kappa score. The experimental results indicate that the stacked ensemble model consistently achieved high performance across both datasets, demonstrating strong robustness and generalization capability.

6.2 TEST CASES



idn	Gender	Age	Hypertension	Heart_disease	Ever_married	Work_type	Residence_type
2e98e30e0e5ffd1f8475255eaa2cee8891bd4ee5	Female	49	0	0	Yes	Private	Urban
60c409717bb1aec593d0411118e2975fb9b6139d	Male	81	0	0	Yes	Private	Urban
0f5888442a92665b65f719dac5712fd287f842b8	Female	80	1	0	Yes	Self-employed	Urban
a080640d17fb8db7f8bf5b0eebb8cf24c8396aae	Female	78	0	0	Yes	Private	Rural
8db2b8c1236bfb85a942a1e45c3c634ad937f34b	Female	68	0	0	No	Govt_job	Urban

Figure 6.2.1: Details of the stroke risk prediction



Model Type	Accuracy
SVM	62.16216216216216
Logistic Regression	62.16216216216216
Decision Tree Classifier	62.16216216216216

7. CONCLUSION & FUTURE ASPECTS

CONCLUSION & FUTURE ASPECTS

In conclusion, the project has successfully achieved its objectives, showcasing significant progress and outcomes. The implementation and execution phases were meticulously planned and executed, leading to substantial improvements and insights. Looking ahead, the future aspects of the project hold immense potential. Future developments will focus on expanding the scope, integrating new technologies, and enhancing sustainability. These advancements will not only strengthen the existing framework but also open new avenues for growth and innovation, ensuring the project remains relevant and impactful in the long term. This strategic approach will drive continuous improvement and success.

7.1 PROJECT CONCLUSION

The proposed system presents an efficient heart disease risk prediction model using a dual-stage stacked machine learning approach. The model integrates multiple machine learning algorithms including Random Forest, Decision Tree, and Extreme Gradient Boosting to improve prediction accuracy and reliability. Hyperparameter tuning techniques such as RandomizedSearchCV and GridSearchCV were applied to obtain the best model parameters and improve overall performance. The final stacked ensemble model achieved a high accuracy of about **96%**, along with strong performance in precision, recall, F1-score, and ROC-AUC metrics.

The model effectively reduces the risk of overfitting by applying k-fold cross-validation, ensuring better generalization on unseen data. Experimental results show that the proposed system provides accurate and reliable predictions with very low false-negative rates, which is crucial in medical diagnosis. The model was also validated using an additional dataset with 1000 instances and achieved an accuracy of 96.88%, proving the stability and robustness of the system. the developed heart disease prediction model can assist healthcare professionals in early detection of cardiovascular diseases and help in making better medical decisions

7.2 FUTURE ASPECTS

In the future, the proposed heart disease prediction system can be improved in several ways. First, advanced **feature selection techniques** can be applied to reduce the number of input attributes while maintaining or improving prediction accuracy. This will also reduce computational complexity and make the model more efficient.

Second, the system can be extended by integrating **deep learning models such as Artificial Neural Networks (ANN)** or hybrid deep learning architectures to capture more complex patterns in medical data. These models may further improve the prediction performance and reliability of the system.

Another future improvement is the use of **larger and more diverse healthcare datasets** collected from different populations. This will help in building a more generalized prediction model capable of detecting multiple types of diseases across different medical domains. Additionally, the system can be developed into a **web or mobile-based clinical decision support system** that allows doctors and patients to perform real-time risk analysis and monitoring.

8. BIBLIOGRAPHY

BIBLIOGRAPHY

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